

How this book was built

Appendices · generated

What this document is

Every word in this book comes from the markdown in this repository: the module READMEs, the two or three teaching notes per module, the labs, the source lists and the shared reference tables. Nothing was rewritten for print — the book *is* the repository, re-typeset so it can be read and skimmed the way a textbook is read, instead of the way hundreds of files are read.

Each **module** is a part. Inside a part you get, in order: the **overview** (the module README, with its time estimate and exit check), the **teaching notes** (`notes/01-...`, `notes/02-...`, and in one case a third), the **lab exercises** with their stated deliverables, any **code templates** that ship as markdown beside that module's scripts, and one chapter of **primary sources and further reading**. Module 16 is special: it opens with the map of the tech stack and is then split into its seven tracks — 16.1 Python through 16.7 Claude Code and the Agent SDK — each a part in its own right.

Reading conventions

Convention	Meaning
Chapter <code>n.m</code>	Printed at the top of every chapter; the contents page, this index and the PDF outline use the same numbering.
Repo path in brackets, e.g. [see also] <code>[08-rag/notes/02-advanced-patterns-and-evaluation.md]</code>	A relative link from the markdown, which print cannot follow. The words stay, and the target path is printed next to them.
Blue links	Real <code>https://</code> URLs remain clickable in the PDF.
Code blocks	Printed verbatim. Long lines wrap, so ASCII diagrams that fit the column keep their alignment and ones that do not will visibly fold.
No source listings	The runnable scripts in each module's <code>code/</code> folder are deliberately not echoed here — they are meant to be run, not read. The lab chapters tell you which ones to run and in what order.
Slide decks	Every module also ships <code>slides/deck.json</code> plus a built <code>.pptx</code> ; those are not part of the book.

How to rebuild it

```
cd <repo root>
.venv/bin/python -m pip install markdown pypdf websockets
.venv/bin/python _tools/build_book.py                # the whole book
.venv/bin/python _tools/build_book.py 04-llm 08-rag    # a subset, for a quick look
```

The build prints the entire book through headless Chrome as one document, so page numbers run continuously from the cover to the last page. It then reads the chapter breaks back out of the printed PDF, which gives both the split points for the per-chapter files and the page numbers printed in the contents. Contents and body therefore cannot disagree.

Build facts

Parts	26
Chapters	123
Source markdown files	160
Printed pages	637
Paper and type	US Letter; 10.4pt Charter body, Helvetica Neue headings, SF Mono code
Generator	<code>_tools/build_book.py</code>